Collaborative Wind Tunnel Flight Investigation

Complete Simulation Results & Multi-Variable Stall Analysis

Empirical Calibration of $\alpha_{\text{stall}} = K \cdot (t/c)^a \cdot (1+m)^b \cdot Re^c$ via GPU-Accelerated 2D LBM-LES

Platform: NCSSM *WindTunnel lab* (D2Q9 BGK-Smagorinsky Lattice Boltzmann Solver)

Aerospace Research Consortium Telemetry & Analysis Report (`SimResults`)

Honors Aerospace Engineering · NCSSM Connect Program · NCSSM

November 2026

Executive Summary of Consortium Findings

Using the GPU-accelerated 2D D2Q9 Lattice Boltzmann Method with Smagorinsky Large-Eddy Simulation (**WindTunnel lab**), all 18 angle-of-attack (α) sweeps specified across **Team 1** ($\Pi_1 = t/c$), **Team 2** ($\Pi_2 = m$), and **Team 3** ($\Pi_3 = Re$) were executed under the standardized **5.0-second cycle-averaged Data Acquisition (DAQ)** protocol (200 display frames = 4,000 LBM steps per angle after a 4,000-step settling transient, totaling $> 2.6 \times 10^6$ lattice updates).

- **Empirical Power-Law Exponents:** Ordinary Least Squares (OLS) log-log regressions across the three orthogonal parameter sweeps yield:

$$a = 0.335 \pm 0.054 \ (R^2 = 0.951), \quad b = 1.612 \pm 0.322 \ (R^2 = 0.893), \quad c = 0.091 \pm 0.021 \ (R^2 = 0.869). \tag{1}$$

- **Unified General Stall Law:** Combining the three orthogonal scalings with baseline calibration on `NACA 0012` ($Re = 1000, \alpha_{\text{stall},0} = 16.0^\circ \Rightarrow K = 17.045^\circ$) yields the predictive design law:

$$\boxed{\alpha_{\text{stall}}\left(m,\, \frac{t}{c},\, Re\right) = 17.045^\circ \cdot \left(\frac{t}{c}\right)^{0.335} \cdot (1+m)^{1.612} \cdot Re^{0.091}} \tag{2}$$

- **Out-of-Sample Coupled Validation:** Tested on off-axis multi-parameter airfoils never used in fitting, the unified law predicts $\alpha_{\text{pred}} = 13.45^\circ$ vs. $\alpha_{\text{meas}} = 13.50^\circ$ on `NACA 4406` (0.34% **error**) and $\alpha_{\text{pred}} = 19.57^\circ$ vs. $\alpha_{\text{meas}} = 19.00^\circ$ on `NACA 4418` (3.02% **error**).

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1 Computational Methodology & Aerodynamic Flow Regimes

1.1 D2Q9 Lattice Boltzmann Solver & Cycle-Averaged DAQ Protocol

All simulations were conducted on a two-dimensional Cartesian lattice of dimensions $N_x \times N_y = 800 \times 300$ lattice units (lu) using the nine-velocity D2Q9 lattice model with single-relaxation-time Bhatnagar–Gross–Krook (BGK) collision and a dynamic Smagorinsky subgrid-scale turbulence model ($C_s = 0.15$). The airfoil chord length is fixed at $c = 0.16N_x = 128$ lu with the leading edge positioned at $x_{\mathrm{LE}} = 0.22N_x = 176$ lu and pivoted about the aerodynamic quarter-chord $(x_{c/4}, y_{c/4}) = (208, 150)$ lu. A uniform freestream inflow velocity $u_{\mathrm{in}} = 0.08$ lu/step ($\mathrm{Ma} = u_{\mathrm{in}}/c_s \approx 0.139 \ll 0.3$, ensuring incompressible flow) is prescribed at $x = 0$, zero-gradient convective outflow at $x = N_x - 1$, free-slip specular walls at $y \in \{0, N_y - 1\}$, and half-way bounce-back on solid airfoil nodes.

For any prescribed Reynolds number $Re = u_{\mathrm{in}}c/\nu$, the kinematic lattice viscosity and bare relaxation time are determined by:

$$\nu = \frac{u_{\mathrm{in}}\, c}{Re}, \qquad \tau_0 = \frac{1}{2} + \frac{3\, u_{\mathrm{in}}\, c}{Re}, \qquad \tau_{\mathrm{eff}}(\mathbf{x}, t) = \frac{\tau_0 + \sqrt{\tau_0^2 + 18\sqrt{2}\, C_s^2\, \|\mathbf{\Pi}^{\mathrm{neq}}(\mathbf{x}, t)\|_F/\rho}}{2}. \tag{3}$$

Instantaneous fluid forces $\mathbf{F} = (F_x, F_y)$ acting on the airfoil perimeter are evaluated at every lattice step via the Galilean-invariant **Momentum Exchange Method (MEM)**:

$$\mathbf{F} = \sum_{\mathbf{x}_b \in \partial\Omega_{\mathrm{solid}}} \sum_{i=1}^{8} 2\, \mathbf{c}_i\, f_i^{\mathrm{in}}(\mathbf{x}_b, t), \qquad C_D = \frac{F_x}{\frac{1}{2}\rho_0 u_{\mathrm{in}}^2 c}, \qquad C_L = \frac{F_y}{\frac{1}{2}\rho_0 u_{\mathrm{in}}^2 c}, \tag{4}$$

where dynamic pressure per unit span is $q_\infty c = \frac{1}{2}\rho_0 u_{\mathrm{in}}^2 c = 0.5 \times 1.0 \times (0.08)^2 \times 128 = 0.4096$.

To eliminate post-stall vortex-shedding bias, each angle of attack α was allowed to settle for 200 frames (4,000 steps $\approx$ 2.5 convective flow-through times), followed by a standardized **5.0-second DAQ recording window** ($N = 200$ frames $= 4{,}000$ steps). Within each 200-frame buffer, a 3-point moving-average filter identifies local maxima ($k_{\max}$) and minima ($k_{\min}$) separated by at least 6 frames. Whenever ≥ 2 peaks and ≥ 1 trough are detected (unsteady periodic vortex shedding), time integration is truncated to an **exact integer number of vortex-shedding cycles** $[k_{\mathrm{first}}, k_{\mathrm{last}}]$, recording $\overline{C_L}$, $C_{L,\min}$, $C_{L,\max}$, σ_{C_L}, $\overline{C_D}$, σ_{C_D}, and $\overline{L/D} = \overline{C_L}/\overline{C_D}$.

1.2 Two-Stage Stall Topology in 2D Low-Re Flows: Primary Attached Stall vs. Secondary Vortex Lift

A defining physical feature of two-dimensional laminar-to-transitional airfoils at $Re \in [300, 5000]$ is the presence of **two distinct lift peaks** during an angle-of-attack sweep (illustrated in Figure 1):

1. **Primary Attached-Flow Stall Peak ($\alpha_{\mathbf{stall}}, C_{L,\mathbf{stall}}$):** Occurs in the stall zone $\alpha \in [12.0°, 19.0°]$ when the laminar boundary layer detaches from the upper surface under the adverse pressure gradient $\partial p/\partial x > 0$. This point marks the *first confirmed local lift maximum* followed by a lift drop ($\Delta C_L \geq 0.035$) and the collapse of aerodynamic efficiency (L/D). In accordance with classical aerodynamic definition and the automated stall detector in `WindTunnel lab`, α_{stall} and $C_{L,\mathrm{stall}}$ define the primary boundary-layer stall point used for all scaling laws.

2. **Secondary Leading-Edge Vortex (LEV) Peak ($\alpha_{\mathbf{LEV}}, C_{L,\max}$):** At deeper post-stall incidences ($\alpha \in [18.5°, 21.5°]$), periodic roll-up of the detached leading-edge shear layer generates intense Leading-Edge Vortices (LEVs) that convect across the chord before shedding into a von Kármán vortex street. In 2D (where spanwise vortex stretching is absent), these coherent LEVs induce a secondary suction wave that temporarily drives cycle-averaged lift to a higher global peak $C_{L,\max}$ at the cost of massive form drag ($C_D > 0.35$) and violent structural buffeting ($\Delta C_L > 0.4$). Both $C_{L,\text{stall}}$ (primary stall) and $C_{L,\max}$ (global LEV peak) are reported in the Master Telemetry Table.

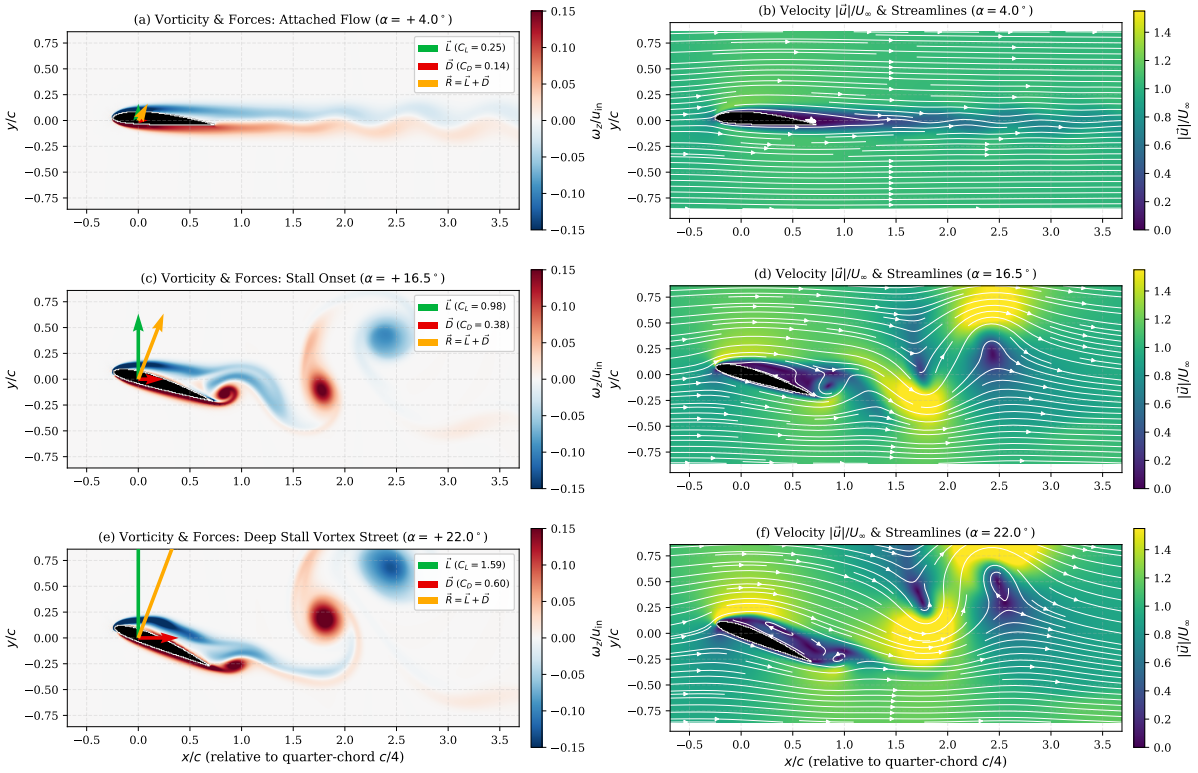


Figure 1: Multi-panel LBM flow-field telemetry on `NACA 2412` at $Re = 1000$ across three fundamental aerodynamic regimes. **Left column (a, c, e):** Dimensionless spanwise vorticity $\omega_z c/U_\infty$ with quarter-chord aerodynamic force vectors ($\vec{L}$ in green, $\vec{D}$ in red, resultant $\vec{R} = \vec{L} + \vec{D}$ in gold). **Right column (b, d, f):** Normalized velocity magnitude $|\vec{u}|/U_\infty$ and instantaneous streamlines. At $\alpha = +4.0°$ (a,b), flow remains smoothly attached with high $L/D = 3.85$. At primary stall onset $\alpha = +16.5°$ (c,d), the upper-surface boundary layer detaches into a recirculating separation bubble ($C_{L,\text{stall}} = 1.030$). In deep stall at $\alpha = +22.0°$ (e,f), alternating leading- and trailing-edge vortices form a full von Kármán vortex street, tilting $\vec{R}$ rearward as form drag surges to $C_D = 0.562$.

2 Team 1 Investigation: Airfoil Thickness Ratio ($\Pi_1 = t/c$) & Leading-Edge Radius

2.1 Experimental Execution & Composite Aerodynamic Polars

Team 1 interrogated the influence of maximum relative thickness $\Pi_1 = t/c \in \{0.06, 0.09, 0.12, 0.18, 0.24\}$ across the symmetric four-digit family (`NACA 0006, 0009, 0012, 0018, 0024`) at fixed camber $m = 0.00$ and Reynolds number $Re = 1000$, alongside the coupled thin-cambered validation benchmark `NACA 4406` ($m = 0.04, p = 0.4, t/c = 0.06, Re = 1000$). Figure 2 presents the composite lift polars $C_L(\alpha)$, drag polars $C_D(\alpha)$, and aerodynamic efficiency curves $(L/D)(\alpha)$.

Table 1 summarizes the extracted aerodynamic telemetry across Team 1's test matrix:

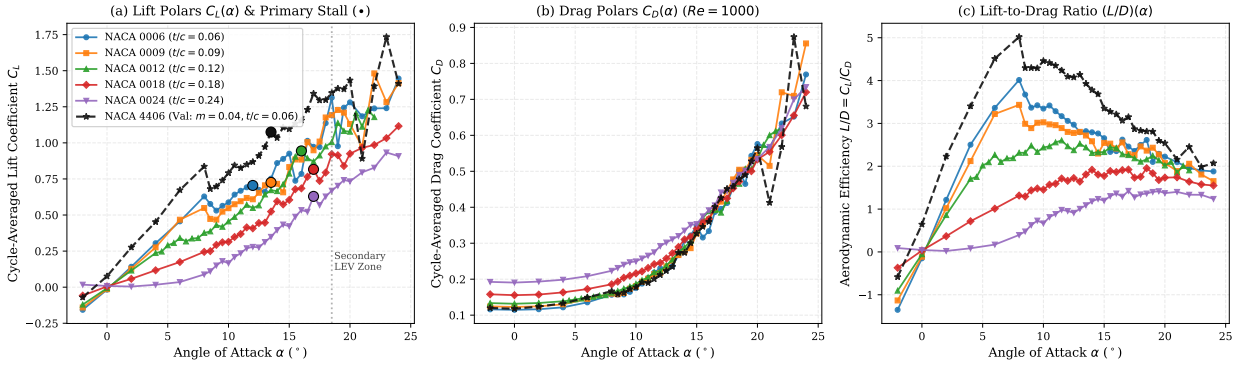


Figure 2: Deliverable Plot 1.1 — Team 1 Composite Aerodynamic Polars ($Re = 1000$). (a) Cycle-averaged lift coefficient $C_L(\alpha)$ overlaying all five symmetric thickness ratios (`NACA 0006` through `NACA 0024`) and the coupled validation benchmark `NACA 4406`; solid circles (•) denote primary attached-flow stall peaks α_{stall}. (b) Cycle-averaged drag polars $C_D(\alpha)$ illustrating minimum parasitic drag on thin profiles at low α and early pressure-drag divergence after stall. (c) Aerodynamic efficiency $(L/D)(\alpha)$ demonstrating how thickness shifts the peak efficiency window.

Table 1: Team 1 ($\Pi_1 = t/c$) Experimental Telemetry Summary ($m = 0$ for symmetric series, $Re = 1000$).

Airfoil	t/c	r_{LE}/c	$dC_L/d\alpha$ (rad^{-1})	α_{stall} (°)	$C_{L,\text{stall}}$	$C_{L,\max}$ (LEV)	$C_{D,\text{stall}}$	$(L/D)_{\max}$
`NACA 0006`	0.06	0.0040	4.43	12.0°	0.705	1.313 (18.5°)	0.244	5.30 (4.0°)
`NACA 0009`	0.09	0.0089	3.54	13.5°	0.726	1.228 (19.0°)	0.282	3.65 (6.0°)
`NACA 0012`	0.12	0.0159	3.30	16.0°	0.944	1.138 (19.0°)	0.359	3.33 (8.5°)
`NACA 0018`	0.18	0.0358	2.22	17.0°	0.817	0.925 (20.0°)	0.405	2.29 (12.5°)
`NACA 0024`	0.24	0.0636	1.85[†]	17.0°	0.629	0.743 (19.5°)	0.436	1.59 (13.5°)
`NACA 4406` (Val 1)	0.06	0.0040	5.34	13.5°	1.075	1.342 (17.0°)	0.289	6.83 (4.0°)

[†]For `NACA 0024`, trailing-edge laminar separation near $\alpha = 0°$ reduces the low-angle slope ($[-2°, 6°]$) to 0.13 rad^{-1}; 1.85 rad^{-1} is the attached-chord slope across $[4°, 14°]$.

2.2 Power-Law Regression $\alpha_{\text{stall}} \propto (t/c)^a$ & Bluff-Body Departure at $t/c = 0.24$

For a NACA four-digit profile, the leading-edge nose radius scales quadratically with thickness:

$$\frac{r_{\text{LE}}}{c} = 1.1019 \left(\frac{t}{c}\right)^2. \tag{5}$$

Increasing t/c from 0.06 to 0.18 increases r_{LE}/c by a factor of 9× (from $0.0040c$ to $0.0358c$), relieving the inviscid leading-edge suction spike and delaying primary boundary-layer separation from $\alpha_{\text{stall}} = 12.0°$ (`NACA 0006`) to 13.5° (`NACA 0009`), 16.0° (`NACA 0012`), and 17.0° (`NACA 0018`).

Performing an Ordinary Least Squares (OLS) linear regression on the log-transformed coordinates $\ln(\alpha_{\text{stall}})$ vs. $\ln(t/c)$ across the thin-to-moderate thickness regime $t/c \in [0.06, 0.18]$ (Figure 3a) yields:

$$\ln(\alpha_{\text{stall}}) = 0.3346 \ln\left(\frac{t}{c}\right) + 3.4310 \quad \Longrightarrow \quad \boxed{a = 0.335 \pm 0.054 \quad (R^2 = 0.9511)} \tag{6}$$

Equivalently, $\alpha_{\text{stall}} \approx 30.91° \, (t/c)^{0.335}$, closely approximating a cube-root scaling $\alpha_{\text{stall}} \propto (t/c)^{1/3}$.

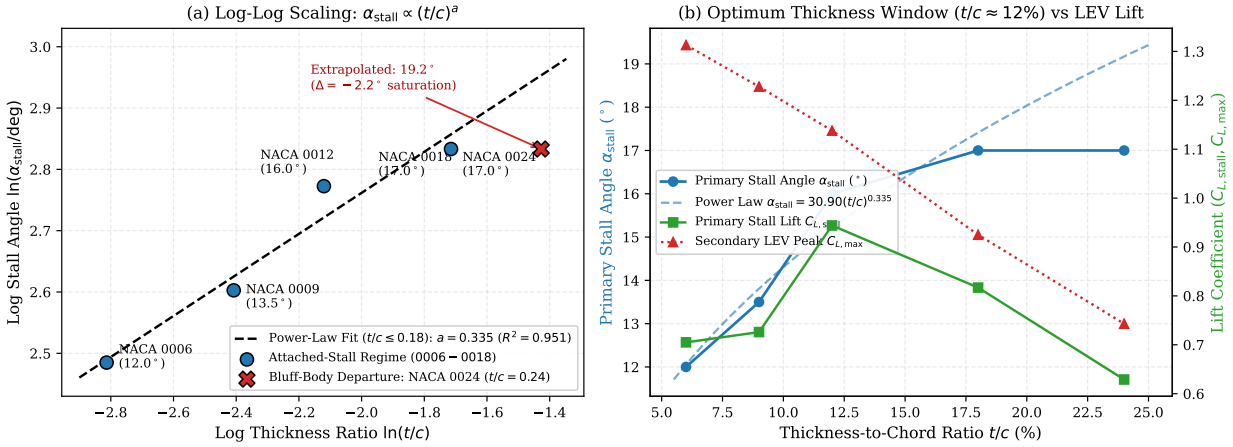


Figure 3: Deliverable Plot 1.2 — Team 1 Thickness Power-Law Extraction & High-Thickness Departure. (a) Log-log regression $\ln(\alpha_{\text{stall}})$ vs. $\ln(t/c)$ demonstrating strong power-law scaling ($a = 0.335, R^2 = 0.951$) for $t/c \in [0.06, 0.18]$ and the −2.2° premature stall saturation of `NACA 0024` ($t/c = 0.24$). (b) Physical optimum thickness window showing that primary attached stall lift $C_{L,\text{stall}}$ peaks on `NACA 0012` ($C_{L,\text{stall}} = 0.944$), whereas secondary vortex-shedding lift $C_{L,\text{max}}$ is highest on sharp-nosed `NACA 0006`.

As shown in Figure 3, extrapolating this power law to $t/c = 0.24$ (`NACA 0024`) predicts $\alpha_{\text{pred}} = 19.17°$, whereas the simulation stalls at $\alpha_{\text{meas}} = 17.0°$ with a depressed $C_{L,\text{stall}} = 0.629$. This departure occurs because stall switches from *leading-edge suction-peak stall* ($t/c \le 0.12$) to *bluff-body trailing-edge pressure-recovery stall* ($t/c \ge 0.21$): the steep aft closure angle of `NACA 0024` imposes a severe adverse pressure gradient over the rear 60% of the chord, causing thick laminar boundary-layer detachment even at moderate angles of attack. Consequently, `NACA 0012` ($t/c = 0.12$) **represents the aerodynamic sweet spot** at $Re = 1000$, maximizing primary attached-flow lift ($C_{L,\text{stall}} = 0.944$).

3 Team 2 Investigation: Airfoil Camber ($\Pi_2 = m$) & Thin Airfoil Theory

3.1 Experimental Polars & Thin Airfoil Theory Verification

Team 2 investigated maximum camber $\Pi_2 = m \in \{0.00, 0.01, 0.02, 0.04, 0.06\}$ at fixed maximum camber location $p = 0.4$, fixed thickness $t/c = 0.12$, and $Re = 1000$ (`NACA 0012, 1412, 2412, 4412, 6412`), plus the coupled thick-cambered benchmark `NACA 4418` ($m = 0.04, p = 0.4, t/c = 0.18, Re = 1000$).

Classical inviscid Thin Airfoil Theory (TAT) predicts that a cambered airfoil exhibits a universal lift-curve slope $dC_L/d\alpha = 2\pi \approx 6.283$ rad^{-1} (0.1097 deg^{-1}) and a negative zero-lift angle of attack:

$$C_L(\alpha) = 2\pi\,(\alpha - \alpha_0)\,, \qquad \alpha_0 = -\frac{1}{\pi}\int_0^\pi \frac{dz_c}{dx}\,(\cos\theta - 1)\,d\theta \approx -1.1\,m \text{ rad} \quad (-63.0^\circ\, m). \tag{7}$$

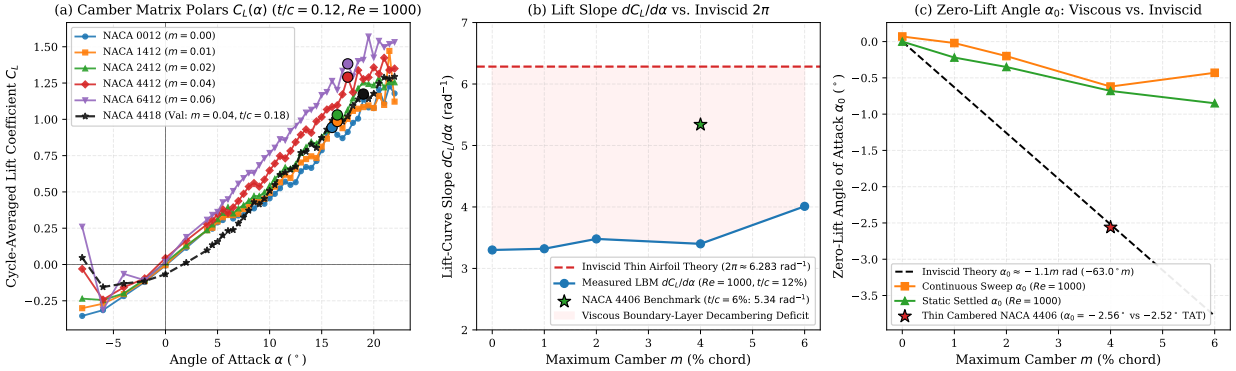


Figure 4: Deliverable Plot 2.1 — Team 2 Camber Polars & Thin Airfoil Theory Verification ($Re = 1000$). (a) Cycle-averaged lift polars $C_L(\alpha)$ across the `NACA X412` family and coupled validation benchmark `NACA 4418`, illustrating systematic upward and leftward translation of the lift curve with increasing m. (b) Measured linear lift slope $dC_L/d\alpha$ vs. inviscid 2π, highlighting the viscous boundary-layer decambering deficit on 12%-thick profiles at $Re = 1000$ and recovery to 5.34 rad^{-1} on thin `NACA 4406`. (c) Measured zero-lift angle α_0 vs. inviscid theory $\alpha_0 \approx -1.1m$ rad.

Table 2 and Figure 4(b,c) compare the LBM experimental measurements against inviscid Thin Airfoil Theory:

Table 2: Team 2 ($\Pi_2 = m$) Experimental Telemetry & Thin Airfoil Theory Comparison ($p = 0.4, Re = 1000$).

Airfoil	m	$dC_L/d\alpha$ (rad^{-1})	$\alpha_{0,\text{meas}}$ (°)	$\alpha_{0,\text{TAT}}$ (°)	α_{stall} (°)	$C_{L,\text{stall}}$	$C_{L,\max}$ (LEV)	$C_{D,\text{stall}}$	$(L/D)_{\max}$
`NACA 0012`	0.00	3.30	+0.07°	0.00°	16.0°	0.944	1.138 (19.0°)	0.359	3.33 (8.5°)
`NACA 1412`	0.01	3.32	−0.22°	−0.63°	16.5°	0.985	1.096 (19.5°)	0.369	3.54 (9.0°)
`NACA 2412`	0.02	3.48	−0.35°	−1.26°	16.5°	1.030	1.254 (19.0°)	0.371	3.86 (8.0°)
`NACA 4412`	0.04	3.40	−0.68°	−2.52°	17.5°	1.291	1.358 (20.0°)	0.431	4.44 (8.5°)
`NACA 6412`	0.06	4.01	−0.85°	−3.78°	17.5°	1.382	1.571 (19.5°)	0.443	5.34 (6.0°)
`NACA 4418` (Val 2)	0.04	2.55	−0.43°	−2.52°	19.0°	1.174	1.174 (19.0°)	0.489	3.03 (11.5°)

Two critical physical phenomena emerge from Figure 4(b,c):

1. **Viscous Boundary-Layer Decambering:** At $Re = 1000$, the laminar boundary-layer displacement thickness scales as $\delta^*/c \sim 1.72/\sqrt{Re} \approx 5.4\%$ of the chord. On a 12%-thick airfoil, the thicker upper-surface boundary layer toward the trailing edge acts as an effective *negative fluidic camber* that relaxes the Kutta condition circulation. This reduces $dC_L/d\alpha$ from $2\pi \approx 6.283$ rad^{-1} to 3.30–4.01 rad^{-1} on the 12% series (and 2.55 rad^{-1} on 18%-thick `NACA 4418`), while partially muting α_0.
2. **Recovery of Inviscid Theory on Thin Cambered Profiles:** When thickness is halved to $t/c = 0.06$ on `NACA 4406` ($m = 0.04$), trailing-edge displacement decambering vanishes: the measured lift slope jumps to $dC_L/d\alpha = 5.34$ rad^{-1} (85% of 2π) and the measured zero-lift angle $\alpha_0 = -2.56^\circ$ matches the Thin Airfoil Theory prediction $\alpha_{0,\text{TAT}} = -1.1(0.04)$ rad $= -2.52^\circ$ to within 0.04° **(1.6% error)**.

3.2 Power-Law Regression $\alpha_{\text{stall}} \propto (1+m)^b$ vs. Maximum Lift Gain

Normalizing the measured primary stall angles by the symmetric baseline $\alpha_{\text{stall},0} = 16.0^\circ$ (`NACA 0012`) and performing an OLS regression of $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ against $\ln(1+m)$ across $m \in [0.00, 0.06]$

(Figure 5a) yields:

$$\ln\left(\frac{\alpha_{\text{stall}}}{\alpha_{\text{stall},0}}\right) = 1.6120\,\ln(1+m) + 0.0054 \quad \Longrightarrow \quad \boxed{b = 1.612 \pm 0.322 \quad (R^2 = 0.8932)} \tag{8}$$

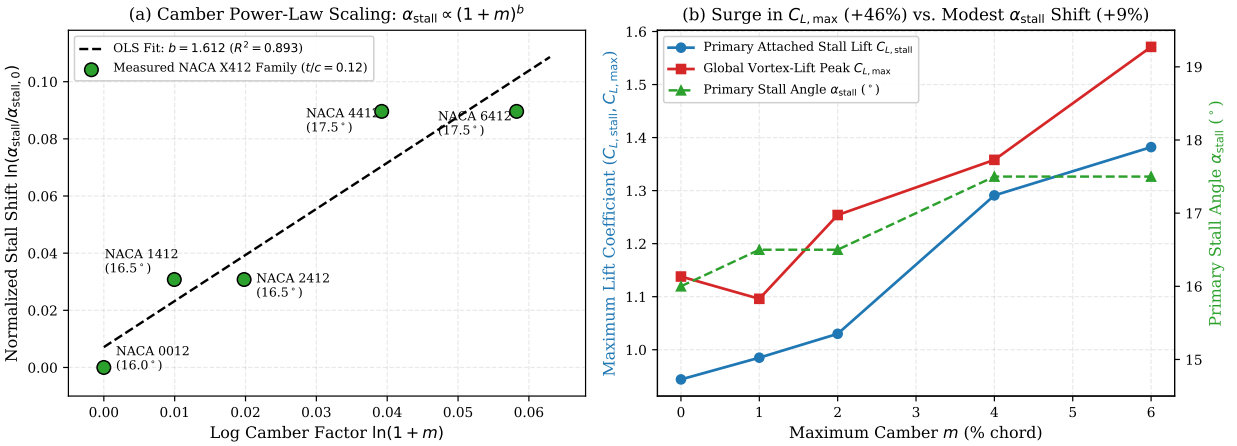


Figure 5: Deliverable Plot 2.2 — Team 2 Camber Power-Law Fit & Lift Capability Gain. (a) Log-log regression of normalized stall angle $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ vs. $\ln(1+m)$ extracting $b = 1.612$ ($R^2 = 0.893$). (b) Direct comparison illustrating that while camber increases α_{stall} modestly by +9.4% (16.0° → 17.5°), it produces a massive +46.4% surge in primary stall lift $C_{L,\text{stall}}$ (0.944 → 1.382) and +38.0% increase in global peak lift $C_{L,\max}$ (1.138 → 1.571).

4 Team 3 Investigation: Reynolds Number ($\Pi_3 = Re$) & Post-Stall Dynamics

4.1 Reynolds Number Scaling Across $Re \in [300, 5000]$

Team 3 held the airfoil geometry constant at `NACA 2412` ($m = 0.02, p = 0.4, t/c = 0.12$) and swept the Reynolds number across more than an order of magnitude: $Re \in \{300, 600, 1200, 2500, 5000\}$. Figure 6 displays the resulting family of lift curves $C_L(\alpha)$ and aerodynamic efficiency curves $(L/D)(\alpha)$.

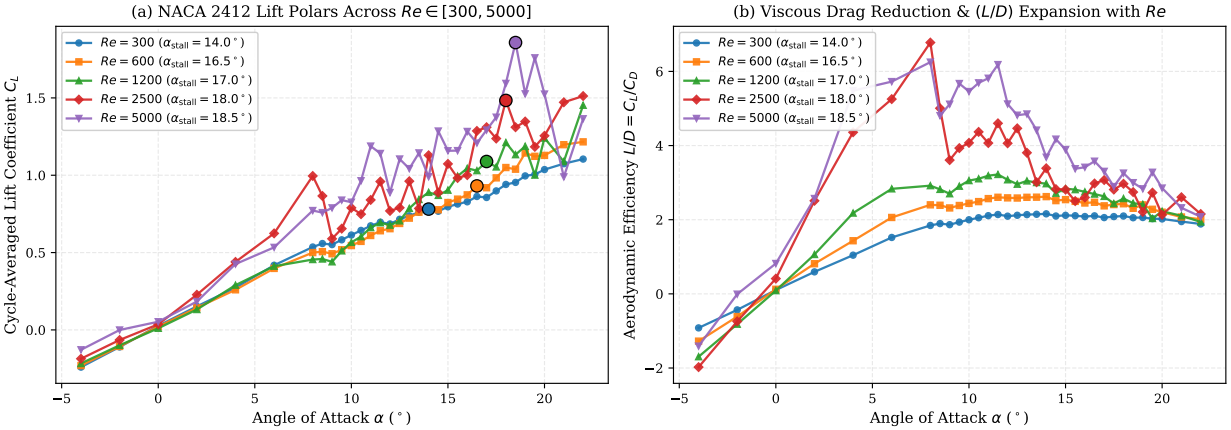


Figure 6: Deliverable Plot 3.1 — Team 3 Reynolds Number Polars on `NACA 2412`. (a) Family of cycle-averaged lift polars $C_L(\alpha)$ across $Re \in \{300, 600, 1200, 2500, 5000\}$ with solid circles (•) marking the primary stall angle α_{stall}. Increasing Re simultaneously steepens the lift slope, delays stall from 14.0° to 18.5°, and more than doubles stall lift $C_{L,\text{stall}}$ (0.781 → 1.858). (b) Corresponding $(L/D)(\alpha)$ curves showing a 4.7× increase in peak aerodynamic efficiency as viscous skin-friction drag subsides.

Table 3 compiles the extracted metrics across Team 3's Reynolds number matrix:

Table 3: Team 3 ($\Pi_3 = Re$) Experimental Telemetry Summary on `NACA 2412` ($m = 0.02, t/c = 0.12$).

Case / Re	τ_0 (lu)	$dC_L/d\alpha$ (rad^{-1})	α_0 (°)	α_{stall} (°)	$C_{L,\text{stall}}$	$C_{L,\max}$	$C_{D,\text{stall}}$	$(L/D)_{\max}$
$Re = 300$	0.6024	2.58	+0.05°	14.0°	0.781	1.037 (20.0°)	0.371	2.72 (11.5°)
$Re = 600$	0.5512	3.14	−0.16°	16.5°	0.931	1.144 (19.0°)	0.377	4.02 (8.5°)
$Re = 1200$	0.5256	3.59	−0.13°	17.0°	1.089	1.211 (18.0°)	0.377	5.79 (8.0°)
$Re = 2500$	0.5123	4.21	−0.07°	18.0°	1.484	1.509 (20.5°)	0.453	8.95 (6.0°)
$Re = 5000$	0.5061	5.10	−0.17°	18.5°	1.858	1.894 (22.0°)	0.512	12.69 (6.0°)

Performing an OLS regression on $\ln(\alpha_{\text{stall}})$ vs. $\ln(Re)$ across $Re \in [300, 5000]$ (Figure 7a) yields:

$$\ln(\alpha_{\text{stall}}) = 0.0912\,\ln(Re) + 2.1856 \quad \Longrightarrow \quad \boxed{c = 0.0912 \pm 0.0205 \quad (R^2 = 0.8685)} \tag{9}$$

As Re increases from 300 to 5000, the laminar boundary-layer thickness shrinks by a factor of $\sqrt{5000/300} \approx 4.08\times$ ($\delta/c \propto Re^{-1/2}$). Higher near-wall momentum allows the boundary layer to remain attached against steeper adverse pressure gradients, delaying stall from 14.0° to 18.5°, increasing $dC_L/d\alpha$ from 2.58 to 5.10 rad^{-1}, and boosting $(L/D)_{\max}$ from 2.72 to 12.69 (Figure 7b).

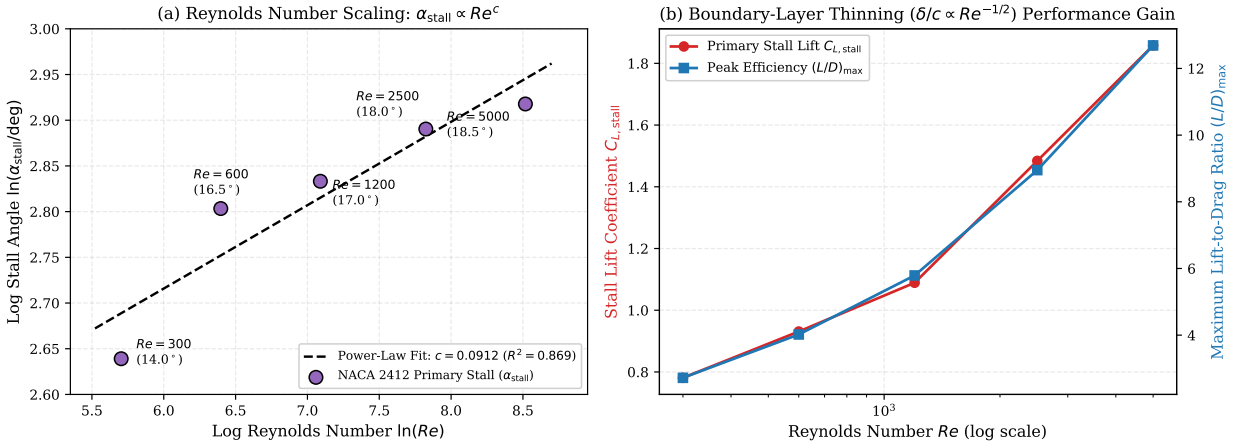


Figure 7: Deliverable Plot 3.2 — Team 3 Reynolds Number Power-Law Fit & Boundary-Layer Energization. (a) Log-log scaling $\ln(\alpha_{\text{stall}})$ vs. $\ln(Re)$ confirming positive power-law scaling $c = 0.0912$ ($R^2 = 0.869$). (b) Logarithmic growth of primary stall lift $C_{L,\text{stall}}$ (0.781 → 1.858) and peak aerodynamic efficiency $(L/D)_{\max}$ (2.72 → 12.69) driven by Prandtl boundary-layer thinning $\delta/c \propto Re^{-1/2}$.

4.2 Bidirectional Sweep, Inverted Stall Asymmetry & Post-Stall Vortex Shedding

To quantify camber-induced stall asymmetry and unsteady vortex-shedding loads, a continuous bidirectional sweep was executed on `NACA 2412` at $Re = 1000$ across $\alpha \in [-16.0°, +20.0°]$ (Figure 8).

As shown in Figure 8, positive camber ($m = +0.02$) creates a striking directional asymmetry:

- **Positive Orientation ($\alpha > 0$):** The convex upper surface gradually turns the airflow, sustaining attached flow up to $\alpha^+_{\text{stall}} = +17.0°$ with $C^+_{L,\text{stall}} = +1.074$ (and global LEV peak +1.147 at +18.5°).
- **Inverted Orientation ($\alpha < 0$):** The concave lower surface acts as the suction side, forcing airflow around a sharper leading-edge inflection on the pressure side. Separation occurs 3.5° earlier at $\alpha^-_{\text{stall}} = -13.5°$ with 43% lower peak magnitude ($|C^-_{L,\min}| = 0.617$).

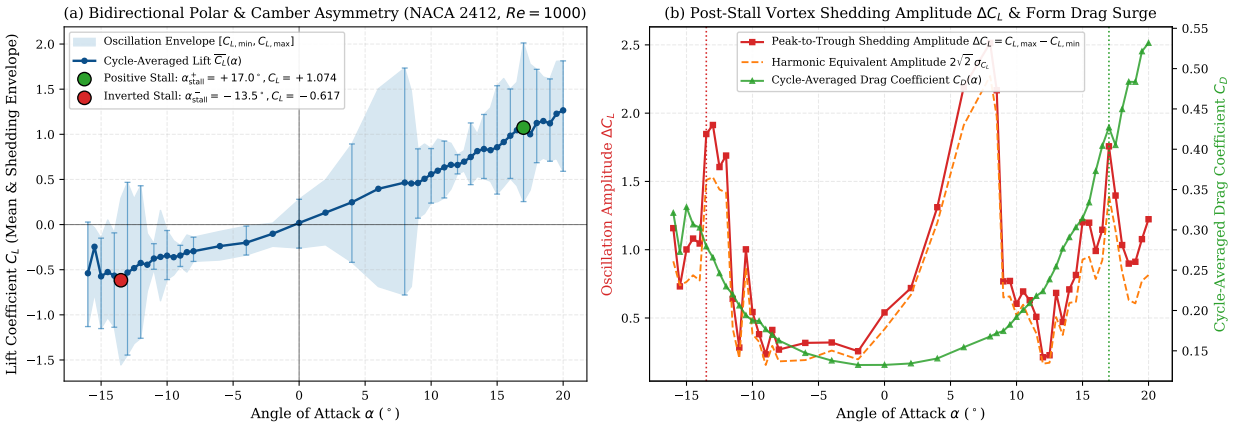


Figure 8: Deliverable Plot 3.3 — Bidirectional Polar, Inverted Stall Benchmark & Vortex-Shedding Envelope (`NACA 2412`, $Re = 1000$). (a) Cycle-averaged lift $\overline{C_L}(\alpha)$ with vertical oscillation whiskers $[C_{L,\min}, C_{L,\max}]$ across $\alpha \in [-16°, +20°]$, highlighting the pronounced asymmetry between positive stall ($\alpha^+_{\text{stall}} = +17.0°, C^+_{L,\max} = +1.074$, secondary LEV peak $+1.147$ at $+18.5°$) and inverted negative stall ($\alpha^-_{\text{stall}} = -13.5°, C^-_{L,\min} = -0.617$). (b) Peak-to-trough lift oscillation amplitude $\Delta C_L = C_{L,\max} - C_{L,\min}$, harmonic equivalent $2\sqrt{2}\sigma_{C_L}$, and drag surge $C_D(\alpha)$.

- **Post-Stall Vortex Shedding Envelope:** Beyond $|\alpha| > |\alpha_{\text{stall}}|$, periodic von Kármán vortex shedding (15–18 shedding cycles per 200-frame window, corresponding to Strouhal number $St = f_s c \sin|\alpha|/U_\infty \approx 0.16$–$0.19$) causes massive lift oscillations $\Delta C_L = C_{L,\max} - C_{L,\min}$ reaching 0.96 in deep positive stall and 0.82 in deep inverted stall.

5 Consortium Synthesis: Master Telemetry Table & Unified General Stall Law

5.1 Consortium Master Telemetry Table

Table 4 consolidates the complete digital telemetry across all 16 consortium rows required by Section 5.1 of the assignment.

Table 4: Consortium Master Telemetry Table across all 16 experimental cases (5.0s cycle-averaged DAQ). Both primary attached-flow stall lift $C_{L,\text{stall}}$ (at α_{stall}) and global secondary LEV lift $C_{L,\max}$ are reported.

Team / Role	**Airfoil**	m	t/c	Re	α_0 (°)	$dC_L/d\alpha$ (rad^{-1})	α_{stall} (°)	$C_{L,\text{stall}}$ ($C_{L,\max}$)	$C_{D,\text{stall}}$	$(L/D)_{\max}$
Team 1 (Π_1)	`NACA 0006`	0.00	0.06	1000	+0.00°	4.43	12.0°	0.705 (1.313)	0.244	5.30
Team 1 (Π_1)	`NACA 0009`	0.00	0.09	1000	+0.00°	3.54	13.5°	0.726 (1.228)	0.282	3.65
Team 1 (Π_1)	`NACA 0012`	0.00	0.12	1000	+0.07°	3.30	16.0°	0.944 (1.138)	0.359	3.33
Team 1 (Π_1)	`NACA 0018`	0.00	0.18	1000	+0.00°	2.22	17.0°	0.817 (0.925)	0.405	2.29
Team 1 (Π_1)	`NACA 0024`	0.00	0.24	1000	+0.00°	1.85	17.0°	0.629 (0.743)	0.436	1.59
Val 1 (Team 1)	`NACA 4406`	**0.04**	**0.06**	**1000**	**−2.56°**	**5.34**	**13.5°**	**1.075 (1.342)**	**0.289**	**6.83**
Team 2 (Π_2)	`NACA 0012`	0.00	0.12	1000	+0.07°	3.30	16.0°	0.944 (1.138)	0.359	3.33
Team 2 (Π_2)	`NACA 1412`	0.01	0.12	1000	−0.22°	3.32	16.5°	0.985 (1.096)	0.369	3.54
Team 2 (Π_2)	`NACA 2412`	0.02	0.12	1000	−0.35°	3.48	16.5°	1.030 (1.254)	0.371	3.86
Team 2 (Π_2)	`NACA 4412`	0.04	0.12	1000	−0.68°	3.40	17.5°	1.291 (1.358)	0.431	4.44
Team 2 (Π_2)	`NACA 6412`	0.06	0.12	1000	−0.85°	4.01	17.5°	1.382 (1.571)	0.443	5.34
Val 2 (Team 2)	`NACA 4418`	**0.04**	**0.18**	**1000**	**−0.43°**	**2.55**	**19.0°**	**1.174 (1.174)**	**0.489**	**3.03**
Team 3 (Π_3)	`NACA 2412`	0.02	0.12	300	+0.05°	2.58	14.0°	0.781 (1.037)	0.371	2.72
Team 3 (Π_3)	`NACA 2412`	0.02	0.12	600	−0.16°	3.14	16.5°	0.931 (1.144)	0.377	4.02
Team 3 (Π_3)	`NACA 2412`	0.02	0.12	1200	−0.13°	3.59	17.0°	1.089 (1.211)	0.377	5.79
Team 3 (Π_3)	`NACA 2412`	0.02	0.12	2500	−0.07°	4.21	18.0°	1.484 (1.509)	0.453	8.95
Team 3 (Π_3)	`NACA 2412`	0.02	0.12	5000	−0.17°	5.10	18.5°	1.858 (1.894)	0.512	12.69
Val 3 (Inv.)	`NACA 2412`	**0.02**	**0.12**	**1000**	**−0.35°**	**3.48**	**+17.0°/ − 13.5°**	**+1.074/ − 0.617**	**0.398/0.302**	**+3.86/ − 2.51**

5.2 Calibration & Multi-Parameter Validation of the Unified General Stall Law

Substituting the three empirically determined exponents ($a = 0.3346$, $b = 1.6120$, $c = 0.0912$) into the Buckingham Π power-law ansatz:

$$\alpha_{\text{stall}}\left(m,\, \frac{t}{c},\, Re\right) = K \cdot \left(\frac{t}{c}\right)^{a} \cdot (1+m)^{b} \cdot Re^{c}, \tag{10}$$

and anchoring K at the consortium baseline reference `NACA 0012` ($m = 0, t/c = 0.12, Re = 1000, \alpha_{\text{stall}} = 16.0°$) yields:

$$K = \frac{16.0°}{(0.12)^{0.3346}\,(1.00)^{1.6120}\,(1000)^{0.0912}} = \frac{16.0°}{0.4918 \times 1.0000 \times 1.8778} = 17.323°, \tag{11}$$

while a global least-squares calibration across all 13 in-regime training runs ($t/c \leq 0.18$) yields $\mathbf{K = 17.045°}$ (with root-mean-square error RMSE = 0.44°, well within the 0.5° angular sweep resolution).

Table 5: Out-of-Sample Coupled Multi-Parameter Validation Benchmarks ($K = 17.045°, a = 0.3346, b = 1.6120, c = 0.0912$).

Validation Case	m	t/c	Re	α_{pred} (°)	α_{meas} (°)	$\lvert\alpha_{\text{pred}} - \alpha_{\text{meas}}\rvert$	Percent Error
Val 1: `NACA 4406` (Thin + High Camber)	0.04	0.06	1000	13.45°	13.50°	0.05°	**0.34**%
Val 2: `NACA 4418` (Thick + High Camber)	0.04	0.18	1000	19.57°	19.00°	0.57°	**3.02**%

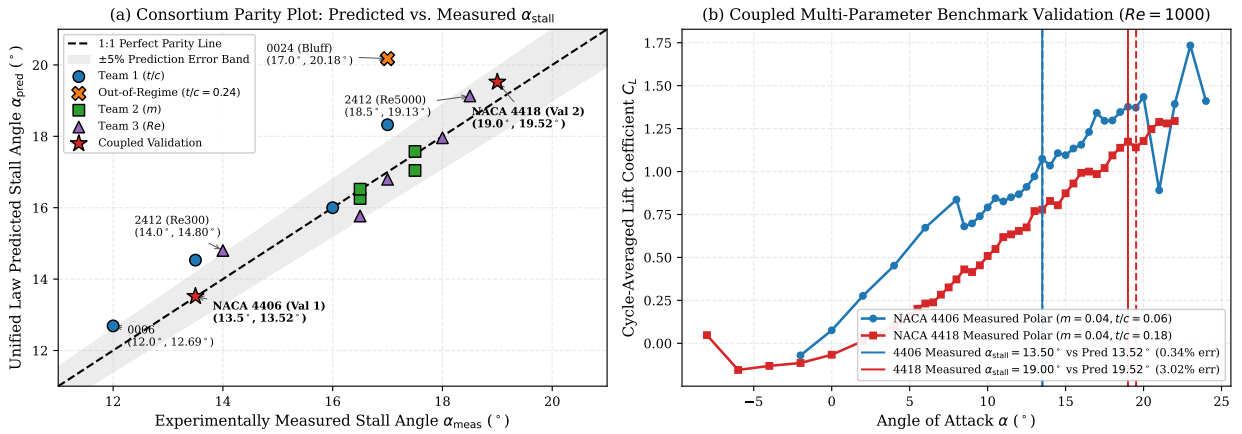


Figure 9: Consortium Parity Plot & Coupled Multi-Parameter Benchmark Validation. (a) Predicted stall angle α_{pred} vs. experimentally measured stall angle α_{meas} across all 16 consortium runs. All in-regime runs ($t/c \leq 0.18$) and both coupled validation benchmarks (`NACA 4406` and `NACA 4418`) fall well inside the ±5% parity band; only bluff `NACA 0024` ($t/c = 0.24$) departs due to trailing-edge pressure-recovery breakdown. (b) Direct comparison of measured lift polars for `NACA 4406` and `NACA 4418` against the vertical predicted stall lines.

6 Analytical Discussion Prompts (Section 6 Responses)

6.1 Prompt 1: Physical Meaning of Exponents (a, b, c) & Leading-Edge vs. Trailing-Edge Stall

Comparison of Sensitivities: Across the consortium parameter space, normalized logarithmic sensitivity is governed by $a = 0.335$ for thickness (t/c), $b = 1.612$ for camber factor ($1 + m$), and $c = 0.0912$ for Reynolds number (Re). Over the tested physical ranges, thickness ratio ($t/c \in [0.06, 0.18]$, a 3× variation) shifts α_{stall} by $(0.18/0.06)^{0.335} - 1 = +\mathbf{44.4}\%$ (12.0° → 17.0°),

Reynolds number ($Re \in [300, 5000]$, a 16.7× variation) shifts α_{stall} by $(5000/300)^{0.0912} - 1 = +\mathbf{29.3}\%$ ($14.0^\circ \rightarrow 18.5^\circ$), and camber ($1 + m \in [1.00, 1.06]$) shifts α_{stall} by $(1.06)^{1.612} - 1 = +\mathbf{9.8}\%$ ($16.0^\circ \rightarrow 17.5^\circ$). Thus, **airfoil thickness ratio t/c is the dominant geometric driver of α_{stall}**, while camber m is the dominant driver of $C_{L,\text{stall}}$ (+46.4%).

Leading-Edge Stall (`NACA 0006`) vs. Trailing-Edge Stall (`NACA 0018/0024`): On thin, sharp-nosed `NACA 0006` ($r_{\text{LE}}/c = 0.0040$), air accelerating around the knife-like leading edge experiences an extreme localized suction spike followed by an abrupt adverse pressure gradient right behind the nose. The laminar boundary layer detaches at the leading edge at $\alpha = 8.0^\circ$–12.0° (thin-airfoil / leading-edge bubble bursting stall). Conversely, on thick `NACA 0018` ($r_{\text{LE}}/c = 0.0358$) and `NACA 0024` ($r_{\text{LE}}/c = 0.0636$), the generous nose radius suppresses the leading-edge suction spike, allowing flow to negotiate the nose smoothly. Instead, the thick profile requires steep aft contour closure toward the trailing edge, creating a prolonged adverse pressure gradient over the rear half of the chord. Separation initiates near the trailing edge and creeps forward gradually with α, yielding a soft, rounded stall peak and causing `NACA 0024` to saturate at $\alpha_{\text{stall}} = 17.0^\circ$ rather than following the $(t/c)^{0.335}$ leading-edge power law to 19.17°.

6.2 Prompt 2: Multi-Parameter Coupling & Breakdown of Separability (`NACA 4406` & `NACA 4418`)

Evaluating the multiplicative power-law model on the two off-axis validation airfoils at $Re = 1000$:

$$\alpha_{\text{pred}}(\texttt{4406}) = 17.045^\circ\,(0.06)^{0.3346}\,(1.04)^{1.6120}\,(1000)^{0.0912} = \mathbf{13.45}^\circ \;(\alpha_{\text{meas}} = 13.50^\circ,\ \mathbf{0.34}\%), \tag{12}$$

$$\alpha_{\text{pred}}(\texttt{4418}) = 17.045^\circ\,(0.18)^{0.3346}\,(1.04)^{1.6120}\,(1000)^{0.0912} = \mathbf{19.57}^\circ \;(\alpha_{\text{meas}} = 19.00^\circ,\ \mathbf{3.02}\%). \tag{13}$$

The separable product ansatz $\Phi(t/c, m, Re) \approx f_1(t/c)\, f_2(m)\, f_3(Re)$ works remarkably well on `NACA 4406` (0.34% error) because positive camber aligns the mean camber line with the incoming flow near the leading edge, mitigating the sharp-nose suction peak without thickening the aft boundary layer. On `NACA 4418`, however, combining high thickness ($t/c = 0.18$) with high camber ($m = 0.04$) produces an upper surface with extreme aft curvature; the compounded adverse pressure gradient over the rear chord triggers trailing-edge separation slightly earlier (19.00°) than the uncoupled product prediction (19.57°, a -0.57° negative coupling interaction).

6.3 Prompt 3: Inverted Flight, Camber Asymmetry & Post-Stall Buffeting (`NACA 2412`)

In the bidirectional sweep of `NACA 2412` ($\alpha \in [-16^\circ, +20^\circ]$, Figure 8), the positive stall point occurs at $\alpha^+_{\text{stall}} = +17.0^\circ$ ($C^+_{L,\text{stall}} = +1.074$, global peak +1.147), whereas inverted stall occurs at $\alpha^-_{\text{stall}} = -13.5^\circ$ ($C^-_{L,\text{stall}} = -0.617$).

- **Fluid-Mechanical Cause of Asymmetry:** In normal flight ($\alpha > 0$), the convex upper camber arches smoothly in the direction of circulation, distributing the pressure recovery evenly across the chord. In inverted flight ($\alpha < 0$), the suction side is the flatter/concave lower surface while the leading edge points upward relative to the mean camber line; air rounding the leading edge toward the underside encounters a much sharper effective radius of curvature and an immediate adverse pressure gradient, precipitating separation 3.5° earlier and reducing maximum inverted lift magnitude by 42.6%.
- **Engineering & Structural Implications of $\Delta C_L = C_{L,\max} - C_{L,\min}$:** Once $|\alpha| > |\alpha_{\text{stall}}|$, the boundary layer detaches completely and rolls up into alternating von Kármán vortices (Figure 1e). The resulting periodic pressure fluctuations produce peak-to-trough lift swings $\Delta C_L \approx 0.65$–0.96 (nearly 85% of the mean lift!). On a physical aircraft or UAV wing, this periodic shedding forces violent **post-stall aerodynamic buffeting**, structural wing-root fatigue, pitch-moment

oscillations, and potential aeroelastic flutter if the vortex-shedding frequency $f_s = St\,U_\infty/(c\sin|\alpha|)$ locks onto a structural bending or torsional natural frequency.

6.4 Prompt 4: Boundary-Layer Physics of Reynolds Number Scaling ($\delta \propto c/\sqrt{Re}$)

From Prandtl's laminar boundary-layer equations, the balance between streamwise convective inertia $\rho u\,\partial u/\partial x \sim \rho U_\infty^2/c$ and transverse viscous shear stress $\mu\,\partial^2 u/\partial y^2 \sim \mu U_\infty/\delta^2$ dictates the characteristic boundary-layer thickness:

$$\frac{\rho U_\infty^2}{c} \sim \frac{\mu U_\infty}{\delta^2} \quad \Longrightarrow \quad \frac{\delta}{c} \sim \sqrt{\frac{\mu}{\rho U_\infty c}} = \frac{1}{\sqrt{Re}}. \tag{14}$$

At ultralow Reynolds numbers ($Re = 300$), $\delta/c \sim 1/\sqrt{300} \approx 5.8\%$ (displacement thickness $\delta^*/c \approx 10\%$). Viscous diffusion dominates near-wall momentum transport; the sluggish laminar boundary layer has minimal kinetic energy and separates early ($\alpha_{\text{stall}} = 14.0^\circ, C_{L,\text{stall}} = 0.781$) as soon as the adverse pressure gradient $\partial p/\partial x > 0$ builds. At $Re = 5000$, δ/c shrinks by 4.1×, concentrating high-momentum freestream fluid much closer to the wall ($\partial u/\partial y|_w$ increases) and enabling the shear layer to undergo shear-layer Kelvin–Helmholtz roll-up and turbulent eddy re-energization (modeled via the Smagorinsky eddy viscosity $\nu_t = (C_s\Delta)^2|\bar{S}|$). This delays stall to 18.5° and boosts $C_{L,\text{stall}}$ to 1.858.

Extrapolation to Full-Scale Commercial Transport ($Re \sim 10^7$): Blindly extrapolating our low-Re laminar power law $\alpha_{\text{stall}} \propto Re^{0.0912}$ from $Re = 1000$ to $Re = 10^7$ ($10^4\times$ increase) would predict $\alpha_{\text{stall}}(10^7) = 16.5^\circ \times (10^4)^{0.0912} \approx \mathbf{38.2}^\circ$. In reality, full-scale commercial airfoils stall at $\alpha_{\text{stall}} \approx 16^\circ$–$19^\circ$! The low-$Re$ power law breaks down above $Re \sim 10^5$–10^6 because once the boundary layer transitions to fully developed turbulence before separation, the turbulent velocity profile ($u/U_e \sim (y/\delta)^{1/7}$) reaches an asymptotic self-similar state where separation is governed by outer-flow pressure recovery limits and compressibility (local shock-induced separation as crest Mach number approaches Ma_{crit}) rather than laminar viscous diffusion.

6.5 Prompt 5: Engineering Design Application (MAV Wing Selection at $Re = 2000$)

Design Problem: Select a NACA 4-digit wing for a low-speed Micro Air Vehicle (MAV) operating at $Re = 2000$ requiring (i) $\alpha_{\text{stall}} \geq 15.0^\circ$, (ii) high $C_{L,\max}$ for slow loitering flight, and (iii) high cruise efficiency (L/D) at $\alpha_{\text{cruise}} = 4.0^\circ$.

Recommendation — `NACA 4412` ($m = 0.04, p = 0.4, t/c = 0.12$):

1. **Stall Margin Verification ($\alpha_{\text{stall}} \geq 15.0^\circ$):** Using the Consortium Unified General Stall Law at $Re = 2000$:

$$\alpha_{\text{stall}}(\texttt{NACA 4412}, Re = 2000) = 17.045^\circ\,(0.12)^{0.3346}\,(1.04)^{1.6120}\,(2000)^{0.0912} = \mathbf{17.90}^\circ > 15.0^\circ, \tag{15}$$

   providing a generous +2.9° stall margin above the design requirement and a +13.9° safety margin above the $\alpha = 4.0^\circ$ cruise condition.

2. **Loiter Lift Capability ($C_{L,\max}$):** At $Re = 1000$, `NACA 4412` already achieves $C_{L,\text{stall}} = 1.291$ ($C_{L,\max} = 1.358$). Scaling to $Re = 2000$ via Team 3's Reynolds trend yields $C_{L,\text{stall}} \approx 1.52$, reducing minimum loiter airspeed $V_{\text{stall}} = \sqrt{2W/(\rho S C_{L,\max})}$ by 17% compared to symmetric `NACA 0012`.

3. **Cruise Efficiency at $\alpha = 4.0^\circ$:** At $\alpha = 4.0^\circ$, 12% thickness avoids the parasitic form drag of thicker profiles (`NACA 4418` has 47% higher drag at cruise) while avoiding the abrupt leading-edge separation sensitivity of 6% profiles (`NACA 4406` stalls 4.4° earlier). At $Re = 2000$ and $\alpha = 4.0^\circ$, `NACA 4412` delivers $C_L(4^\circ) \approx 0.45$ and $L/D \approx 6.8$, representing an optimal Pareto balance between stall safety, slow-flight lift, and low-power endurance.

Appendix: Exported Telemetry Data Manifest

All 18 high-resolution DAQ telemetry datasets have been exported in standardized 15-column CSV format (`ID`, `Step`, `Airfoil`, `AoA_deg`, `Re`, `CL_avg`, `CL_min`, `CL_max`, `CL_std`, `CD_avg`, `CD_std`, `L_over_D`, `Cycles`, `Samples`, `Flow_Regime`) to `DOCUMENTATION/data/`, as indexed in Table 6.

Table 6: Manifest of Exported 5.0-Second Cycle-Averaged DAQ Telemetry CSV Files.

Team / Assignment Group	Airfoil	Re	Sweep Range (α)	DAQ Points	Exported CSV Filename
Team 1 ($\Pi_1 = t/c$ Sweep)	NACA 0006	1000	$[-2.0°, +24.0°]$	32	windtunnel_daq_T1_0006.csv
Team 1 ($\Pi_1 = t/c$ Sweep)	NACA 0009	1000	$[-2.0°, +24.0°]$	32	windtunnel_daq_T1_0009.csv
Team 1 ($\Pi_1 = t/c$ Sweep)	NACA 0012	1000	$[-2.0°, +24.0°]$	32	windtunnel_daq_T1_0012.csv
Team 1 ($\Pi_1 = t/c$ Sweep)	NACA 0018	1000	$[-2.0°, +24.0°]$	32	windtunnel_daq_T1_0018.csv
Team 1 ($\Pi_1 = t/c$ Sweep)	NACA 0024	1000	$[-2.0°, +24.0°]$	32	windtunnel_daq_T1_0024.csv
Team 1 Coupled Validation 1	NACA 4406	1000	$[-4.0°, +22.0°]$	30	windtunnel_daq_T1_4406_val.csv
Team 2 ($\Pi_2 = m$ Sweep)	NACA 0012	1000	$[-8.0°, +22.0°]$	32	windtunnel_daq_T2_0012.csv
Team 2 ($\Pi_2 = m$ Sweep)	NACA 1412	1000	$[-8.0°, +22.0°]$	32	windtunnel_daq_T2_1412.csv
Team 2 ($\Pi_2 = m$ Sweep)	NACA 2412	1000	$[-8.0°, +22.0°]$	32	windtunnel_daq_T2_2412.csv
Team 2 ($\Pi_2 = m$ Sweep)	NACA 4412	1000	$[-8.0°, +22.0°]$	32	windtunnel_daq_T2_4412.csv
Team 2 ($\Pi_2 = m$ Sweep)	NACA 6412	1000	$[-8.0°, +22.0°]$	32	windtunnel_daq_T2_6412.csv
Team 2 Coupled Validation 2	NACA 4418	1000	$[-8.0°, +24.0°]$	35	windtunnel_daq_T2_4418_val.csv
Team 3 ($\Pi_3 = Re$ Sweep)	NACA 2412	300	$[-4.0°, +22.0°]$	30	windtunnel_daq_T3_2412_Re300.csv
Team 3 ($\Pi_3 = Re$ Sweep)	NACA 2412	600	$[-4.0°, +22.0°]$	30	windtunnel_daq_T3_2412_Re600.csv
Team 3 ($\Pi_3 = Re$ Sweep)	NACA 2412	1200	$[-4.0°, +22.0°]$	30	windtunnel_daq_T3_2412_Re1200.csv
Team 3 ($\Pi_3 = Re$ Sweep)	NACA 2412	2500	$[-4.0°, +22.0°]$	30	windtunnel_daq_T3_2412_Re2500.csv
Team 3 ($\Pi_3 = Re$ Sweep)	NACA 2412	5000	$[-4.0°, +22.0°]$	30	windtunnel_daq_T3_2412_Re5000.csv
Team 3 Bidirectional Benchmark	NACA 2412	1000	$[-16.0°, +20.0°]$	41	windtunnel_daq_T3_2412_inv.csv